

Linear algebra: Exercise Final Review

Morteza Rahimi

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1. Consider the following matrix.

$$A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

- Compute eigenvalues and corresponding eigenvectors of A .
- Determine algebraic and geometric multiplicity of eigenvalues of A .
- Is A diagonalizable? If so, diagonalize it ($A = X\Lambda X^{-1}$).
- What is the spectral decomposition of $A = Q\Lambda Q^T$.
- Compute the singular values and left and right singular vectors of A .
- Find singular value decomposition of $A = U\Sigma V^T$.
- Is it positive definite? Why?
- Compute determinant and trace of A using eigenvalues.
- Compute A^4 .
- Compute A^{-1} using spectral decomposition of A .
- Compute the singular values and left and right singular vectors of A .
- Compute a matrix that is similar to A .

2. Consider the following matrix.

$$B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

- Compute the singular values and left and right singular vectors of B .
- Calculate singular value decomposition of $B = U\Sigma V^T$.
- Compute bases for $C(B)$, $N(B^T)$, $C(B^T)$, and $N(B)$.
- Show that row space $C(B^T)$ and null space $N(B)$ are orthogonal.
- Are U and V positive definite? If so, provide reasoning; otherwise, which type of matrices are they?

3. Consider the following vectors and spaces.

$$a = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad S_1 = \text{span}\{a, b\}, \quad S_2 = \text{span}\{a, b, c\}.$$

- Compute the orthogonal projection of a onto the line through b . Also, find the projection matrix P .
 - Compute the orthogonal projection of c onto the subspace S_1 . Also, find the projection matrix P .
 - Compute an orthogonal base for space S_2 .
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